SYNTHETIC DEMONSTRATION DATA. NOT VALID FOR REAL MACHINE OPERATION.

Standard Operating Procedure: Spindle & Coolant Controls

Page 1: Section 1: General Machine Overview

This document defines the basic startup and warmup routine for DEMO-CNC-01 3-axis machining center. All operators must hold Level 1 Machining Certification.

WARNING: This is Revision A, originally released 2021-03-01. Superseded by Revision B on 2024-01-15. Do not use for current operations.

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Page 2: Section 2: Spindle Warm-up Routine (Legacy)

Spindle Warm-up Procedure (Rev A Legacy):
Execute a single 15-minute warmup sequence at constant 1,000 RPM. Verify spindle chiller oil temperature reads between 20°C and 25°C.

Legacy Coolant Parameters: Maintain water-soluble synthetic coolant at 5% to 7% Brix concentration. Flood coolant pressure set to 1.5 bar.

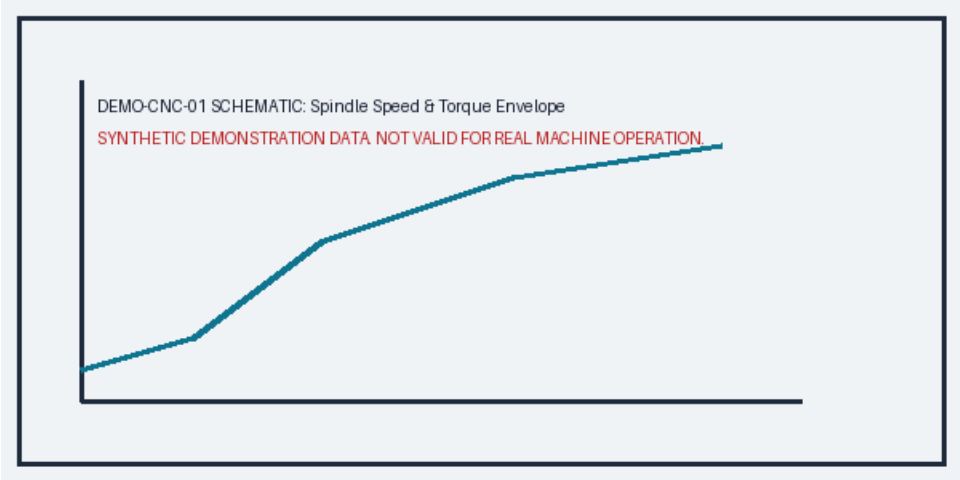


Figure 1: SOP-CNC-001 Operational Limits Diagram

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Standard Operating Procedure: Spindle & Coolant Controls

Page 3: Section 3: Legacy Cutting Parameters

Feed Rate Guidelines:
For general roughing on 6061-T6 Aluminum, feed rate is 2,000 mm/min at 8,000 RPM.
For Titanium Grade 5 (Ti-6Al-4V), maximum recommended feed rate is 1,500 mm/min.

Safety Note: Ensure door interlock switches are engaged before cycle start.